science fact?

So far, teleportation experiments have involved just single photons or electrons. To teleport a living human being, the computer would need information about every single particle in every single atom of the human body. One mistake, and it could be a dead body that Scotty beams aboard!

is allowed to interact with an electron, and all the information about this interaction (which destroys the original quantum state of the photon) is stored. The interaction will have changed the quantum state of the photon in a box on the Moon, but nobody on the Moon knows this yet. Now, the results of the interaction between the electron and the photon back on Earth can be sent to the Moon, by rocket, or laser beam, or any conventional method that does not involve traveling faster than light. Armed with this information, a skilled physicist on the Moon could tweak the photon in the box in such a way that the tweaking subtracts out the changes caused by the entanglement, and produces an exact copy of the first photon. By any conceivable test it is the first photon.

It should be emphasized that this really has been done, for pairs of photons separated by several feet. At one level, this is a somewhat futile thing to do for a photon, because the whole process takes place much slower than light speed, and it would be quicker to send the photon across the room in the conventional manner. But it demonstrates that, in principle, it is possible for an exact copy of a physical

American-born physicist **David Bohm** (1917–92) was one of the first people to appreciate the essentially nonlocal nature of quantum physics. His ideas were largely ignored from the 1950s to the 1980s, but were rehabilitated thanks to John Bell and Alain Aspect. They are at the forefront of 21st century thinking about practical applications of quantum phenomena.

key points

- Quantum physics is not just a wild idea. It has been tested and proved accurate to many decimal places
- Quantum computers will make a single laptop more powerful than every present-day computer on Earth put together
- Single photons have already been "teleported" across 6.2 miles (10 km)

object to be transported in this way any distance across space, with the condition that the whole process must take place at less than light speed.

taxing quantum matter

Just how soon any of these ideas become practicable in the world outside the laboratory remains to be seen, but there can be little doubt that they will affect our lives in the not-too-distant future. More than 150 years ago, not long after he had invented the electric motor, Michael Faraday was asked by a leading politician of the day what use his invention was. He replied to the effect that he had no idea what practical use his invention might be put to, but that he was sure politicians would find a way to tax it.

"I think I can safely say that nobody understands quantum mechanics...do not keep saying to yourself, if you can possibly avoid it, 'but how can it be like that?' because you will go 'down the drain' into a blind alley from which nobody has yet escaped. Nobody knows how it can be like that."

Richard Feynman, *The Character of Physical Law*, BBC (1965)

It surely won't be too long before taxes are being paid on the income generated by quantum computers, quantum cryptography, quantum teleportation, and other as yet undreamed-of devices. Almost instantaneous travel, genuinely intelligent computers, and perfect "virtual realities" are probably just around the corner. As Arthur C. Clarke said, "any sufficiently advanced technology is indistinguishable from magic." We will have magic at our fingertips within a hundred years.

an unimaginable future

The electric motor started out as a toy. Even Michael Faraday, who invented it, could not predict that it would lead to trains that traveled at 217mph (350kph).

new horizons

Before the invention of the steam engine, the human world had scarcely changed for thousands of years. Now, the power of the quantum is opening the door to an era of unprecedented change and unimaginable possibilities.